

Theorem 5.6 (Schwarz reflection principle) *Suppose that f is a holomorphic function in Ω^+ that extends continuously to I and such that f is real-valued on I . Then there exists a function F holomorphic in all of Ω such that $F = f$ on Ω^+ .*

Proof. The idea is simply to define $F(z)$ for $z \in \Omega^-$ by

$$F(z) = \overline{f(\bar{z})}.$$

To prove that F is holomorphic in Ω^- we note that if $z, z_0 \in \Omega^-$, then $\bar{z}, \bar{z}_0 \in \Omega^+$ and hence, the power series expansion of f near $\bar{z}_0$ gives

$$f(\bar{z}) = \sum a_n(\bar{z} - \bar{z}_0)^n.$$

As a consequence we see that

$$F(z) = \sum \bar{a}_n(z - z_0)^n$$

and F is holomorphic in Ω^- . Since f is real valued on I we have $\overline{f(x)} = f(x)$ whenever $x \in I$ and hence F extends continuously up to I . The proof is complete once we invoke the symmetry principle.

5.5 Runge's approximation theorem

We know by Weierstrass's theorem that any continuous function on a compact interval can be approximated uniformly by polynomials.⁴ With this result in mind, one may inquire about similar approximations in complex analysis. More precisely, we ask the following question: what conditions on a compact set $K \subset \mathbb{C}$ guarantee that any function holomorphic in a neighborhood of this set can be approximated uniformly by polynomials on K ?

An example of this is provided by power series expansions. We recall that if f is a holomorphic function in a disc D , then it has a power series expansion $f(z) = \sum_{n=0}^{\infty} a_n z^n$ that converges uniformly on every compact set $K \subset D$. By taking partial sums of this series, we conclude that f can be approximated uniformly by polynomials on any compact subset of D .

In general, however, some condition on K must be imposed, as we see by considering the function $f(z) = 1/z$ on the unit circle $K = C$. Indeed, recall that $\int_C f(z) dz = 2\pi i$, and if p is any polynomial, then Cauchy's theorem implies $\int_C p(z) dz = 0$, and this quickly leads to a contradiction.

⁴A proof may be found in Section 1.8, Chapter 5, of Book I.

A restriction on K that guarantees the approximation pertains to the topology of its complement: K^c must be connected. In fact, a slight modification of the above example when $f(z) = 1/z$ proves that this condition on K is also necessary; see Problem 4.

Conversely, uniform approximations exist when K^c is connected, and this result follows from a theorem of Runge which states that for *any* K a uniform approximation exists by *rational functions* with “singularities” in the complement of K .⁵ This result is remarkable since rational functions are globally defined, while f is given only in a neighborhood of K . In particular, f could be defined independently on different components of K , making the conclusion of the theorem even more striking.

Theorem 5.7 *Any function holomorphic in a neighborhood of a compact set K can be approximated uniformly on K by rational functions whose singularities are in K^c .*

If K^c is connected, any function holomorphic in a neighborhood of K can be approximated uniformly on K by polynomials.

We shall see how the second part of the theorem follows from the first: when K^c is connected, one can “push” the singularities to infinity thereby transforming the rational functions into polynomials.

The key to the theorem lies in an integral representation formula that is a simple consequence of the Cauchy integral formula applied to a square.

Lemma 5.8 *Suppose f is holomorphic in an open set Ω , and $K \subset \Omega$ is compact. Then, there exists finitely many segments $\gamma_1, \dots, \gamma_N$ in $\Omega - K$ such that*

$$(15) \quad f(z) = \sum_{n=1}^N \frac{1}{2\pi i} \int_{\gamma_n} \frac{f(\zeta)}{\zeta - z} d\zeta \quad \text{for all } z \in K.$$

Proof. Let $d = c \cdot d(K, \Omega^c)$, where c is any constant $< 1/\sqrt{2}$, and consider a grid formed by (solid) squares with sides parallel to the axis and of length d .

We let $\mathcal{Q} = \{Q_1, \dots, Q_M\}$ denote the finite collection of squares in this grid that intersect K , with the boundary of each square given the positive orientation. (We denote by ∂Q_m the boundary of the square Q_m .) Finally, we let $\gamma_1, \dots, \gamma_N$ denote the sides of squares in $\mathcal{Q}$ that do not belong to two adjacent squares in $\mathcal{Q}$. (See Figure 13.) The choice of d guarantees that for each n , $\gamma_n \subset \Omega$, and γ_n does not intersect K ; for if it did, then it would belong to two adjacent squares in $\mathcal{Q}$, contradicting our choice of γ_n .

⁵These singularities are points where the function is not holomorphic, and are “poles”, as defined in the next chapter.

Since γ does not intersect K , the integrand $F(z, t)$ in this last integral is jointly continuous on $K \times [0, 1]$, and since K is compact, given $\epsilon > 0$, there exists $\delta > 0$ such that

$$\sup_{z \in K} |F(z, t_1) - F(z, t_2)| < \epsilon \quad \text{whenever } |t_1 - t_2| < \delta.$$

Arguing as in the proof of Theorem 5.4, we see that the Riemann sums of the integral $\int_0^1 F(z, t) dt$ approximate it uniformly on K . Since each of these Riemann sums is a rational function with singularities on γ , the lemma is proved.

Finally, the process of pushing the poles to infinity is accomplished by using the fact that K^c is connected. Since any rational function whose only singularity is at the point z_0 is a polynomial in $1/(z - z_0)$, it suffices to establish the next lemma to complete the proof of Theorem 5.7.

Lemma 5.10 *If K^c is connected and $z_0 \notin K$, then the function $1/(z - z_0)$ can be approximated uniformly on K by polynomials.*

Proof. First, we choose a point z_1 that is outside a large open disc D centered at the origin and which contains K . Then

$$\frac{1}{z - z_1} = -\frac{1}{z_1} \frac{1}{1 - z/z_1} = \sum_{n=1}^{\infty} -\frac{z^n}{z_1^{n+1}},$$

where the series converges uniformly for $z \in K$. The partial sums of this series are polynomials that provide a uniform approximation to $1/(z - z_1)$ on K . In particular, this implies that any power $1/(z - z_1)^k$ can also be approximated uniformly on K by polynomials.

It now suffices to prove that $1/(z - z_0)$ can be approximated uniformly on K by polynomials in $1/(z - z_1)$. To do so, we use the fact that K^c is connected to travel from z_0 to the point z_1 . Let γ be a curve in K^c that is parametrized by $\gamma(t)$ on $[0, 1]$, and such that $\gamma(0) = z_0$ and $\gamma(1) = z_1$. If we let $\rho = \frac{1}{2}d(K, \gamma)$, then $\rho > 0$ since γ and K are compact. We then choose a sequence of points $\{w_1, \dots, w_\ell\}$ on γ such that $w_0 = z_0$, $w_\ell = z_1$, and $|w_j - w_{j+1}| < \rho$ for all $0 \leq j < \ell$.

We claim that if w is a point on γ , and w' any other point with $|w - w'| < \rho$, then $1/(z - w)$ can be approximated uniformly on K by polynomials in $1/(z - w')$. To see this, note that

$$\begin{aligned} \frac{1}{z - w} &= \frac{1}{z - w'} \frac{1}{1 - \frac{w - w'}{z - w'}} \\ &= \sum_{n=0}^{\infty} \frac{(w - w')^n}{(z - w')^{n+1}}, \end{aligned}$$

and since the sum converges uniformly for $z \in K$, the approximation by partial sums proves our claim.

This result allows us to travel from z_0 to z_1 through the finite sequence $\{w_j\}$ to find that $1/(z - z_0)$ can be approximated uniformly on K by polynomials in $1/(z - z_1)$. This concludes the proof of the lemma, and also that of the theorem.

6 Exercises

1. Prove that

$$\int_0^\infty \sin(x^2) dx = \int_0^\infty \cos(x^2) dx = \frac{\sqrt{2\pi}}{4}.$$

These are the **Fresnel integrals**. Here, $\int_0^\infty$ is interpreted as $\lim_{R \rightarrow \infty} \int_0^R$.

[Hint: Integrate the function e^{-z^2} over the path in Figure 14. Recall that $\int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi}$.]

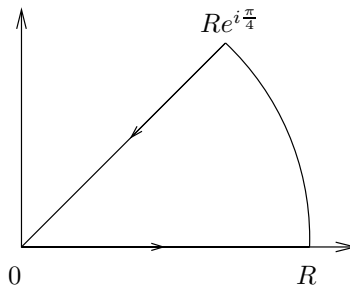


Figure 14. The contour in Exercise 1

2. Show that $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$.

[Hint: The integral equals $\frac{1}{2i} \int_{-\infty}^\infty \frac{e^{ix} - 1}{x} dx$. Use the indented semicircle.]

3. Evaluate the integrals

$$\int_0^\infty e^{-ax} \cos bx dx \quad \text{and} \quad \int_0^\infty e^{-ax} \sin bx dx, \quad a > 0$$

by integrating e^{-Az} , $A = \sqrt{a^2 + b^2}$, over an appropriate sector with angle ω , with $\cos \omega = a/A$.